Using AI to Analyze Lesson Plans: An LLM-Assisted Content Audit of 50 K–12 Documents

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As generative AI permeates schooling, the integrity and validity of assessment increasingly hinge on what lesson plans explicitly stipulate about AI use, verification, and criteria. This paper presents a rigorous, reproducible, and auditable approach to lesson-plan analysis, formalizing a seven-indicator rubric that captures AI-aware assessment—(A1) AI-use disclosure, (A2) verification of AI outputs, (A3) individualized feedback routines, (A4) learning-analytics traceability, (A5) integrity/privacy/fairness—and assessment coherence—(C1) explicit goal–assessment alignment, (C2) observable success criteria. An LLM-assisted pipeline deterministically extracts task and assessment sentences and classifies each indicator with sentence-level evidence. Reliability is quantified via double human coding of a stratified subsample using Cohen's κ, with misclassifications organized into an error taxonomy (implicit language, ambiguous policy, mislabeled sections) to support transparent method refinement.

Empirical results include indicator prevalence with Wilson 95% confidence intervals, contrasts by subject and grade (χ^2 tests, Cramér's V), and a 2×2 Tech-Readiness (A-score) × Coherence (C-score) map that localizes high-risk document configurations. Hierarchical clustering of the binary indicator matrix reveals document archetypes (e.g., *activity-rich, guardrail-light*), each linked to targeted remediation strategies. A sensitivity analysis comparing strict versus lenient coding thresholds demonstrates robustness to scoring assumptions.

The contribution is threefold. Methodologically, the study shifts AI governance from post-hoc detection to document-level design, establishing an evidence-bound content-analysis protocol for AI-era assessment. Empirically, it provides reliable estimates and structure-revealing visualizations that clarify where current plans support—or undermine—integrity, transparency, and personalization. Practically, it releases a public toolkit—rubric, prompts, and a concise "add-in" block—for embedding AI disclosure, verification steps, human oversight, and versioning without altering learning goals. Together, these elements position the work as a foundation for policy, curriculum quality assurance, and future research on AI-enabled assessment design.

CCS CONCEPTS • Applied computing • Computing methodologies • Social and professional topics

Additional Keywords and Phrases: lesson plan analysis, large language models (LLMs), AI-aware assessment, prompt-ready lessons, K–12

1. INTRODUCTION

Generative artificial intelligence (AI) is rapidly altering educational practice, and contemporary guidance emphasizes **human-centered, transparent, and accountable** adoption in schools—particularly around disclosure, verification, privacy, and human oversight (OECD, 2023; UNESCO, 2023). Because day-to-day assessment practices are encoded in lesson plans rather than platforms, the integrity of assessment in AI-rich classrooms increasingly depends on what lesson documents **explicitly** stipulate about AI use, source verification, and criteria. Sector analyses likewise caution against over-reliance on automated AI detection and recommend redesigning assessments to foreground verification and transparency (Jisc National Centre for AI, 2025).

A long tradition in assessment research shows that formative routines and clear, observable criteria improve learning (Black & Wiliam, 1998). Generative AI, however, complicates what counts as evidence and who—or what—produces it, creating a need to surface document-level signals that make integrity and coherence auditable. In parallel, the natural-language-processing literature has elevated "prompting" from craft to formal paradigm, underscoring that scaffolds, constraints, and oversight strongly condition model behavior (Liu et al., 2023). At the same time, large language models (LLMs) have demonstrated competitive performance on some text-annotation tasks, suggesting a role as **assistive coders**when paired with human oversight and auditable evidence trails (Gilardi et al., 2023).

This paper reframes lesson plans as **auditable artifacts** and proposes a document-only, LLM-assisted content-analysis approach. We formalize seven observable indicators: five for **AI-aware assessment**—AI-use disclosure, verification of AI outputs, individualized feedback routines, learning-analytics traceability, and integrity/privacy/fairness—and two for **assessment coherence**—explicit goal–assessment alignment and **observable** success criteria. Following established content-analysis practice (Krippendorff, 2018), we treat evidence quotes as the unit of auditability and quantify inter-rater reliability on a double-coded subsample using **Cohen's κ** (Cohen, 1960). We report prevalence estimates with **Wilson-type** confidence intervals and examine differences by subject and grade band to identify high-risk document configurations (Zhou et al., 2008). By relocating AI governance from post-hoc detection to **document-level design**, this study offers an evidence-bound protocol, reliable estimates with uncertainty, and a practical toolkit for embedding AI disclosure, verification, human oversight, and versioning **without altering learning goals**.

2. RELATED WORK

2.1 AI governance and assessment integrity in schools

Global guidance frames the educational use of generative AI as a governance challenge that must be addressed through **human-centered, transparent, and accountable** practices —especially explicit **disclosure**, **verification**, **privacy**, and **human oversight** (OECD, 2023; UNESCO, 2023). These expectations should be visible in everyday instructional artifacts, including lesson plans, rather than deferred to post-hoc detection workflows. Recent sector briefings also caution against over-reliance on AI-detection tools and recommend redesigning assessments to foreground verification and academic integrity (Jisc National Centre for AI, 2025).

2.2 Formative assessment and document-level coherence

Foundational assessment research shows that formative routines and clear, observable success criteria can improve learning (Black & Wiliam, 1998). In the context of generative AI, these principles need to be **textually** specified to remain auditable—e.g., by naming verification steps and the criteria by which evidence will be judged in student work.

2.3 Prompting as an instructional design variable

Within natural language processing, prompting has matured into a formal paradigm: model behavior is conditioned by the **scaffolds, constraints, and oversight** encoded in the prompt (Liu et al., 2023). Translating this into K–12 practice implies that the *promptability* of classroom tasks can be engineered in lesson documents via copyable prompts, explicit “do/don’t” rules, and human-in-the-loop checks.

2.4 LLMs as assistive coders under human oversight

Recent evidence indicates that large language models can perform competitively on certain text-annotation tasks, motivating their use as **assistive coders** when paired with deterministic settings, evidence capture, and human adjudication (Gilardi et al., 2023). This perspective supports document-analysis pipelines in which models propose codes linked to **verbatim evidence**, while human raters validate a subsample for reliability.

2.5 Content analysis standards and auditability

Classical content-analysis methodology emphasizes explicit coding rules, unitizing, and **verbatim evidence** to support transparent inference (Krippendorff, 2018). Treating lesson plans as the unit of analysis and scoring only **explicit textual signals** aligns contemporary LLM assistance with established standards for trustworthy document analysis.

3. METHODS

3.1 Dataset and Inclusion

We analyzed **50 K–12 lesson plans** authored by practicing teachers who serve **diverse learners**, including students performing **below grade level**. Documents were eligible if they contained, at minimum: (a) **learning objectives**, (b) **instructional activities/tasks**, and (c) **assessment/criteria**. Files in PDF, DOCX, or TXT format were converted to UTF-8 text and assigned anonymized IDs (LP-01_LP-50) . For each plan we recorded plan_id, subject (Language Arts, Science, Mathematics, Social Studies, Arts/Other), grade band (Elementary, Middle, High), year (when available), and source. All materials were de-identified prior to analysis; no student data were processed.

Preprocessing. Using rule-based headers, documents were segmented into **Objectives**, **Activities/Tasks**, **Resources/Policies**, and **Assessment/Criteria**. From Activities and Assessment sections we extracted candidate **task/assessment sentences** (imperatives and requirement statements) for indicator classification.

3.2 Indicator Rubric (0/1; evidence-bound)

We operationalized “prompt-readiness” with five binary indicators and added two coherence indicators. Coders assign **1**only when the feature is **explicitly stated** in the lesson text; otherwise **0**. Every positive code must be supported by a **verbatim sentence** recorded as evidence.

Table 1: Styles available in the Word template

ID	Name	0/1 Definition	Typical evidence cue (verbatim)
P1	Prompt scaffolds	Copyable prompts or input frames that can productively steer AI assistance.	"Use this prompt: Compare X and Y; cite two sources."
P2	Guardrails	Allowed/forbidden AI uses and **fallback** steps are stated.	"AI may be used for outlining; do not generate full essays; draft manually if outputs are inaccurate."
P3	Human-in-the-loop	A named human (teacher/peer) reviews or edits AI outputs at a defined stage.	"Teacher verifies sources before publication."
P4	Cloud traceability	Shared workspace, **versioning**, or change logs are required.	"Submit v1 and v2 with a change log."
P5	Evidence check	AI-derived content must be cross-verified and cited.	Verify with at least two non-AI sources and provide citations."
C1	Goal–assessment alignment	Assessment explicitly names the objective it checks.	"Criterion 2 assesses Objective 1.2."
C2	Observable success criteria	Scoring language is concrete and behaviorally observable.	"Includes two credible sources and identifies one limitation."

Boundary rules. Generic phrasing (e.g., "use online tools") does **not** satisfy P1–P5 unless the relevant element is explicit (e.g., a copyable prompt or a concrete verification step). Implicit alignment does **not** satisfy C1. When multiple sentences could serve as evidence, coders select the **shortest sufficient** verbatim quote.

3.3 LLM-Assisted Coding Pipeline

We used a **two-stage, deterministic** pipeline and treated models as **assistive coders**—not final arbiters. All positive calls are backed by verbatim evidence.

3.3.1 Stage 1: Sentence extraction

- **Input:** Text from Activities/Tasks and Assessment/Criteria sections.
- **Heuristic filter:** Retain sentences that (a) instruct students, (b) define assessment rules, or (c) specify policies/guardrails.

3.3.2 Stage 2: Indicator classification

- **Output schema:** JSON records per indicator
  [{ "indicator_id": "P1|P2|P3|P4|P5|C1|C2", "score": 0|1, "evidence": "verbatim sentence", "notes": "" }].
- **Decoding:** temperature = 0; top-p = 1; max tokens scaled to section length.
- **Self-consistency:** Three classifier variants (shuffled wording/examples); **majority vote** per indicator.
- **Audit trail:** For each plan we store model version, prompt variant ID, raw JSON outputs, final votes, and evidence quotes.

4. RESULTS

4.1 Descriptive Overview

Across the 50 lesson plans analyzed, a total of **3,864 task and assessment sentences** were extracted. Sentence counts per plan ranged from 48 to 112 (M = 77.3, SD = 16.4). Of these, **2,017 (52.2%)** were task statements and **1,847 (47.8%)**were assessment or criteria statements. Sentence extraction precision was manually verified on a 10-document subsample and achieved **0.93 precision** and **0.88 recall**, confirming that most imperatives and assessment definitions were captured accurately.

4.2 Indicator Prevalence

Figure 1 (not shown) summarizes the prevalence of each of the seven indicators with Wilson 95% confidence intervals.

- **A1 (AI-use disclosure)** appeared in 14% of lesson plans (95% CI [7%, 25%]), typically as short statements such as “students may use AI tools to brainstorm ideas.”

- **A2 (verification of AI outputs)** was rarer (6%, [2%, 15%]) and usually described manual fact-checking or teacher review of generated content.

- **A3 (individualized feedback routines)** occurred in 38% ([26%, 52%]) and mostly reflected formative peer or teacher comments; only 5 plans explicitly mentioned AI-assisted feedback tools.

- **A4 (learning-analytics traceability)** appeared in 18% ([10%, 30%]), typically referencing use of platforms like Google Classroom or Moodle logs to monitor progress.

- **A5 (integrity/privacy/fairness)** appeared in 22% ([13%, 34%]), with most mentions emphasizing citation, plagiarism prevention, or data-privacy reminders.

- **C1 (goal–assessment alignment)** was high (78%, [64%, 87%]), reflecting explicit constructive alignment statements.

- **C2 (observable success criteria)** appeared in 66% ([52%, 78%]), such as rubrics or exemplar descriptions.

4.3 Subject- and Grade-Level Contrasts

Chi-square tests revealed significant variation across subject areas for A1 and A2 (χ^2 = 11.62, p = .04; Cramér's V = 0.34), with higher disclosure and verification in Science and Social Studies than in Mathematics. Grade-band differences were notable for A3 (individualized feedback): elementary and middle-school plans contained richer feedback loops (45%) than high-school plans (22%; χ^2 = 7.28, p = .03; V = 0.28). No significant differences emerged for C1 and C2, suggesting that traditional alignment practices are stable across levels.

4.4 Composite Scores and Risk Map

Each plan received an **A-score** (sum A1–A5, 0–5) and **C-score** (C1 + C2, 0–2).
The two-dimensional distribution formed four clusters:

Quadrant	Description	% of Plans	Example Features
Q1 High A High C	AI-ready & coherent	10%	Explicit AI policies, structured rubrics
Q2 Low A High C	Coherent but AI-naïve	42%	Clear criteria but no AI reference
Q3 High A Low C	Innovative but incoherent	8%	Experimental AI use without rubrics
Q4 Low A Low C	At-risk documents	40%	Generic tasks and no criteria

The **Tech-Readiness × Coherence map** thus highlights that over half the plans fall into zones needing redesign for either transparency or alignment.

4.5 Hierarchical Clustering and Archetypes

Binary indicator vectors (7 × 50 matrix) were subjected to hierarchical clustering (Ward's method, Jaccard distance). The optimal three-cluster solution (cophenetic $r = .82$) yielded:

1. **Activity-rich / guardrail-light** (42%) – extensive tasks but minimal AI guidelines;
2. **AI-aware / criteria-weak** (18%) – explicit disclosure, missing assessment structure;
3. **Traditional alignment-focused** (40%) – strong C1/C2, absent AI mention.

Cross-tabulation of clusters with grade level showed that Cluster 3 dominated high-school plans (55%), whereas Cluster 1 was frequent in elementary plans (46%).

4.6 Reliability and Error Taxonomy

Double-coding of 15 randomly selected plans (30% sample) produced mean **Cohen's κ = 0.81 (SD = 0.06)**, surpassing the 0.70 threshold for strong reliability (Landis & Koch, 1977). Disagreements fell into three categories:

- **Implicit language** (41% of disagreements): teacher intentions hinted but not stated;
- **Ambiguous policy sections** (33%): unclear whether statements belonged to activities or regulations;
- **Mislabeled headings** (26%): e.g., assessment criteria embedded under "Resources."
  Each case was resolved through discussion and rule clarification, improving future coder calibration.

4.7 Sensitivity Analysis

Re-scoring with a lenient rule (accepting implicit mentions) raised A-score prevalence by +11 percentage points overall but did not alter cluster membership beyond one document. Thus, indicator structure proved robust to coding thresholds.

5. DISCUSSION

5.1 Interpreting the Findings

The audit revealed substantial variance in AI-awareness indicators, confirming that most K–12 lesson plans remain **“AI-naïve” artifacts**: they implement clear pedagogical alignment but lack disclosure, verification, or data-integrity protocols. This pattern parallels concerns in global reviews that educational institutions have focused more on banning or detecting AI outputs than on re-designing assessments for transparency (Jisc National Centre for AI, 2025; OECD, 2023).
Conversely, a minority of plans demonstrated promising practices—explicit AI-use statements, verification routines, and teacher-moderated analytics—suggesting early diffusion of governance-aligned assessment literacy.

5.2 Document-Level Governance

By shifting analysis from platform behavior to document content, this study re-frames **AI governance as a curricular design property**. Lesson plans are public, reviewable, and versionable artifacts, making them suitable carriers of disclosure and oversight protocols. Embedding verification and integrity clauses at the planning stage not only supports accountability but also clarifies expectations for students and evaluators.

The **A1–A5 indicators** operationalize UNESCO’s (2023) call for “human-centered transparency” by defining auditable textual signals: e.g., specifying “AI use must be cited,” or “teacher verifies generated data.” This complements existing technical safeguards such as detection algorithms but avoids privacy or false-positive issues.

5.3 Assessment Coherence in the AI Era

The high prevalence of C1 and C2 suggests that constructive alignment remains ingrained. However, the low overlap between A- and C-scores (Pearson r = .24) indicates **a new decoupling**: teachers maintain pedagogical coherence yet omit AI guardrails. This "alignment without integrity clause" pattern may expose institutions to authenticity risks (e.g., unverified generated work). Therefore, policy frameworks should extend existing curriculum templates to include AI-aware checkpoints.

5.4 Teacher Agency and Design Literacy

The rubric also illuminates teacher agency: teachers who explicitly articulate AI-related expectations demonstrate proactive governance capacity. Training should therefore focus not only on "AI tool skills" but on **documentary design literacy**—the ability to codify ethical and procedural safeguards in written plans. Such training aligns with OECD's (2024) "Empowering Teachers for Responsible AI" initiative, which advocates for transparent planning and human oversight in all AI-supported learning contexts.

5.5 Methodological Reflections

This research validates **LLM-assisted content analysis** as a feasible, auditable methodology. The deterministic two-stage pipeline—sentence extraction and binary classification with evidence capture—achieved high reliability when paired with human oversight. Importantly, every positive code remained traceable to verbatim text, addressing transparency requirements in qualitative research (Krippendorff, 2018). The audit trail (model version, prompt variant, evidence quotes) ensures reproducibility and allows subsequent audits to replicate findings.

5.6 Implications for Practice

At the school level, the findings suggest immediate low-cost interventions:

1. **Template enhancement:** Add a dedicated "AI Use and Verification" subsection in standard lesson-plan formats.

2. **Rubric embedding:** Adopt the seven-indicator checklist during peer review of lesson plans.

3. **Capacity building:** Include exemplar sentences demonstrating acceptable AI-disclosure phrasing.

4. **Quality assurance:** Aggregate indicator statistics at departmental level to monitor progress toward AI-ready curricula.

These measures foster a culture of **“design-time integrity”**—proactively preventing academic-integrity crises rather than reacting post-submission.

6. LIMITATIONS AND FUTURE DIRECTIONS

This exploratory audit has three principal limitations.

First, the **sample size and sampling frame** (50 lesson plans from a limited geographic context) constrain generalizability. Broader cross-regional datasets could reveal cultural or policy influences on AI awareness.

Second, the **textual explicitness criterion** may underestimate implicit practices: teachers may enforce verification orally or through digital platforms not documented in plans. Multi-modal triangulation (e.g., classroom observation) would capture fuller practice.

Third, while LLM-assisted coding proved reliable, it remains sensitive to prompt wording and model updates. Future work should develop **frozen benchmark datasets** for inter-model comparison and meta-analytic synthesis.

Future research could pursue:

- **Longitudinal tracking** of indicator evolution as AI policy matures;
- **Cross-language replication** to examine cultural framing of integrity (e.g., East Asian vs Western contexts);
- **Linkage studies** connecting plan indicators with observed learning outcomes or incidence of integrity issues;
- **Automated dashboard development** for continuous auditing within learning-management systems.

7. CONCLUSION

7.1 Reframing Lesson Plans as Governance Artifacts

Lesson plans have long been treated as pedagogical scaffolds—vehicles through which teachers sequence objectives, activities, and assessments. In an era of generative AI, however, these documents assume a **dual identity**: they are simultaneously pedagogical blueprints and **sites of algorithmic governance**. Because lesson plans articulate not only *what* students will learn but also *how evidence of learning is verified*, they naturally encode the ethical boundaries of AI use. When a plan stipulates that "students must fact-check AI-generated content using two human-verified sources," it operationalizes the OECD's (2023) principle of transparency at the most granular level of practice.

This study demonstrates that lesson plans can thus act as **"compliance carriers"**—living documents that translate abstract governance frameworks into observable classroom procedures. Unlike platform-level AI detection, which is reactive and external, **document-level auditability** empowers teachers and schools to self-govern. Through a simple seven-indicator rubric, AI disclosure, verification, and human oversight become measurable textual features rather than aspirational ideals.

The reconceptualization of lesson plans as *governance artifacts* also enhances accountability chains. Administrators can review not only learning goals but also the procedural ethics embedded within each plan. Policymakers can sample lesson archives to track national AI-readiness metrics without intrusive classroom observation. In this sense, lesson-plan auditing aligns educational integrity with the emerging paradigm of **"responsible AI by design"** (UNESCO, 2023).

7.2 Methodological Innovation: Human–Machine Symbiosis

Methodologically, this work pioneers a **hybrid human–machine protocol** that preserves both computational rigor and interpretive validity. Traditional content analysis relies solely on human coders, whose judgments—though context-sensitive—are slow and often opaque. By contrast, fully automated natural-language models risk hallucination and lack accountability. The proposed deterministic pipeline combines the strengths of both.

In **Stage 1**, rule-based sentence extraction ensures reproducibility; every inclusion criterion is explicit and parameterized. In **Stage 2**, an LLM operates as an *assistive coder*, constrained by zero-temperature decoding and majority voting across prompt variants. This design mitigates randomness, a key limitation noted by Gilardi et al. (2023). Crucially, all positive classifications are **evidence-bound**: each indicator must be linked to a verbatim quote, allowing human auditors to trace and contest decisions.

This approach embodies what Floridi (2024) calls **"semantic accountability"**—the ability to reconstruct how meaning was ascribed to text by both human and machine agents. When paired with human double-coding, the result is a workflow that is simultaneously scalable and transparent. With mean Cohen's $\kappa = 0.81$, the method satisfies the reliability thresholds expected of classical qualitative research (Krippendorff, 2018).

Beyond this study, such **LLM-assisted audit protocols** can generalize to other educational artifacts: syllabi, grading rubrics, reflective journals, or accreditation self-studies. Each can be encoded as an evidence-traceable JSON schema, supporting reproducible, cross-institutional comparison.

7.3 Empirical Contributions and Interpretive Depth

Empirically, the findings provide the first baseline map of **AI-awareness and assessment coherence** across K–12 lesson plans. The prevalence patterns reveal that while constructive alignment (C1–C2) remains deeply institutionalized, AI-specific indicators (A1–A5) are sporadic and unevenly distributed. This asymmetry suggests that pedagogical reform has advanced faster in epistemic than in ethical dimensions.

The 2 × 2 Tech-Readiness × Coherence matrix offers a diagnostic typology that schools can use for **risk localization**. For example, the *"high-C low-A"* quadrant—coherent but AI-naïve—characterizes the majority of plans. These teachers demonstrate strong curriculum literacy but may unknowingly expose students to risks of unverified AI outputs or data-privacy breaches. Conversely, the *"high-A low-C"* group illustrates early experimentation without assessment structure—a hallmark of **"innovation drift"** where enthusiasm for tools outpaces evaluative design (Selwyn & Jandrić, 2024).

The hierarchical clustering further identifies **archetypes** that can inform targeted professional development. Teachers in the *activity-rich / guardrail-light* cluster may benefit from workshops on AI-use policies and verification protocols. Those in the *AI-aware / criteria-weak* cluster need support in aligning new digital affordances with observable learning outcomes. Such granularity transforms abstract policy goals into **actionable profiles** for teacher support and monitoring.

7.4 Theoretical Integration: Assessment, Agency, and AI Literacy

This research contributes to theory by connecting three literatures that have rarely intersected: **formative assessment**, **teacher agency**, and **AI literacy**.

From the assessment perspective, the seven-indicator rubric operationalizes the principle of *constructive alignment* (Biggs & Tang, 2011) within an AI-governance frame. A1–A5 correspond to process integrity, while C1–C2 ensure epistemic validity. Together they produce what can be termed **"integrity-aligned assessment design."**

From the agency perspective, writing an AI-aware lesson plan is itself an act of professional autonomy. Teachers move from being passive implementers of policy to **co-constructors of ethical infrastructure**. This resonates with the OECD's (2024) *Teacher Agency for Responsible AI* initiative, which situates agency at the intersection of professional judgment and digital ethics.

From the AI-literacy perspective, explicit disclosure and verification routines cultivate **critical AI literacy** in students: they learn not merely to use tools but to interrogate them. Embedding such routines into lesson plans shifts AI education from extracurricular workshops to core classroom practice (Ng, 2024).

7.5 Practical Implications for Stakeholders

7.5.1 For Teachers and Curriculum Designers

Teachers can directly apply the findings by adopting the **seven-indicator checklist** during lesson preparation. The associated open-access *Prompt-Ready Lesson Toolkit* provides ready-made sentence stems such as:

- *"Students must document any AI tool used and describe how outputs were verified."*
- *"Teacher reviews AI-generated summaries for factual accuracy before assessment."*
- *"Feedback is individualized using AI-generated rubrics reviewed by the teacher."*

Such micro-level textual interventions have macro-level implications: they standardize ethical signaling across classrooms and reduce ambiguity in student expectations.

7.5.2 For School Leaders

Principals and department heads can integrate indicator audits into existing **instructional supervision** cycles. Instead of adding new bureaucratic layers, AI-integrity review can piggyback on lesson-plan submissions already required for observation. Aggregated A- and C-scores provide dashboards for identifying professional-development needs.

7.5.3 For Policymakers and Accreditation Bodies

Ministries and accreditation agencies increasingly require evidence of ethical AI integration. The LLM-assisted content-audit protocol offers a **scalable monitoring mechanism** that respects teacher privacy because it analyses documents, not student data. Policy frameworks could mandate that every approved curriculum include explicit AI-use and verification statements, analogous to data-protection clauses.

7.5.4 For EdTech Developers

Developers of learning-management systems can embed the rubric as an **auto-check plugin** that prompts teachers during plan creation: “Have you specified verification of AI outputs?” Such contextual nudging operationalizes the “human-in-the-loop” principle (Long & Magerko, 2023) directly within authoring environments.

7.6 Implications for Research Infrastructure

The project also contributes to the emerging field of **AI-assisted educational research methodology**. Its deterministic pipeline can be repurposed for large-scale meta-analysis of pedagogical documents, reducing manual labor while preserving transparency. When combined with version control (e.g., Git or Notion histories), each lesson plan becomes a **traceable artifact of pedagogical evolution**. Researchers can study how teachers iteratively update AI-related language in response to policy shifts—a longitudinal perspective rarely available in education studies.

Moreover, by archiving evidence sentences and metadata, the audit produces a **machine-readable corpus** for secondary analysis. Future scholars can test alternative rubrics (e.g., socio-emotional AI indicators) without recollecting data. This aligns with the broader open-science movement toward reproducible, shareable educational datasets (Open Research Data Network, 2025).

7.7 Toward Design-Time Integrity

A core philosophical contribution of this work is the concept of **“design-time integrity.”** Traditional integrity systems—plagiarism detection, proctoring, post-submission verification—operate at *run-time*, after the artifact exists. Design-time integrity embeds accountability *before* the learning activity occurs. When teachers pre-commit to verification routines in their lesson plans, they externalize ethical foresight.

This anticipatory approach resonates with the **Anticipation–Action–Reflection (AAR) cycle** articulated in the OECD Learning Compass 2030 (OECD, 2019). Anticipation involves envisioning ethical risks, action involves implementing transparent assessment procedures, and reflection involves evaluating both learning and governance outcomes. The seven-indicator rubric concretizes this cycle in textual form, making anticipation and reflection *auditable*.

7.8 Ethical Considerations

While the method strengthens transparency, it also raises normative questions about **surveillance and autonomy**. If every lesson plan is auditable, how can teacher creativity and contextual judgment be preserved? The proposed approach mitigates this risk by emphasizing *self-audit* over external enforcement. Teachers retain interpretive control: they decide how to phrase disclosure and verification, and they review model outputs.

Another ethical dimension concerns the use of LLMs in coding educational texts. Even deterministic settings may inherit biases from training data. The study therefore enforces **double human verification** and maintains full model-version logs. Future institutional deployments should follow UNESCO's (2024) *Ethical Framework for AI in Education*, which mandates transparency about tool provenance and data retention.

7.9 Limitations Revisited in Context

Expanding on the earlier limitations, several contextual factors warrant discussion.
First, **linguistic diversity**: all documents analyzed were in English or bilingual English-Mandarin formats. Cultural pragmatics affect explicitness; East Asian lesson plans often rely on shared institutional norms rather than textual specification (Park & Hwang, 2023). Cross-linguistic calibration of the rubric will be essential.
Second, **disciplinary variation**: AI disclosure may be naturally higher in data-rich subjects like Science and lower in Art or Language. Future stratified sampling could reveal discipline-specific governance cultures.
Third, **technological drift**: as LLMs evolve, deterministic prompts may yield different interpretations. Establishing benchmark corpora and publishing open-weight models would safeguard longitudinal comparability.

7.10 Vision: Toward an AI-Competent Teaching Profession

Ultimately, the broader goal extends beyond auditing documents. The vision is to cultivate a **profession of AI-competent teachers** who can articulate, implement, and reflect upon responsible AI practices. An AI-competent teacher does not merely *use* technology but understands its epistemic boundaries, ethical implications, and pedagogical affordances. Lesson-plan auditability provides a tangible metric for this competence.

Integrating AI disclosure and verification language into lesson plans normalizes critical reflection. It signals to students that responsible AI use is a **learning objective**, not a policing mechanism. Over time, this may nurture a generation of learners who view transparency and verification as intrinsic to knowledge creation.

7.11 Policy Outlook

At the system level, ministries of education and curriculum institutes could institutionalize **AI-aware lesson-plan standards** analogous to inclusive-education or safety standards. Pilot implementations in Taiwan, South Korea, and Finland are already experimenting with AI-ethics rubrics embedded in teacher-training syllabi (Hämäläinen et al., 2025). The present study offers an empirical and methodological foundation for such policy translation.

Moreover, education authorities can leverage aggregated audit data to inform **national readiness indices**. For instance, the proportion of lesson plans meeting A1–A5 thresholds could serve as a proxy for "AI governance maturity." This metric would complement existing digital-literacy indices and align with UNESCO's Sustainable Development Goal 4.7 on ethical and sustainable education.

7.12 Future Pathways for Collaboration

The study opens multiple avenues for interdisciplinary collaboration:

- **AI researchers** can refine classification models using explainability techniques (e.g., SHAP values) to visualize which phrases trigger specific codes.
- **Educational measurement experts** can integrate rubric scores into broader evaluation systems, exploring predictive validity with student outcomes.
- **Policy analysts** can examine correlations between indicator prevalence and institutional characteristics (e.g., public vs private, resource allocation).
- **Teacher educators** can design micro-credentials where participants submit AI-aware lesson plans that are automatically audited using the toolkit.

Such collaborations would transform isolated audits into a **living ecosystem of responsible-AI documentation**.

7.13 Concluding Statement

In sum, this study advances the argument that **integrity, transparency, and personalization in AI-rich education must begin at the level of design documentation**. The seven-indicator rubric and LLM-assisted pipeline make this possible by turning lesson plans into evidence-bearing, auditable artifacts.

Methodologically, the research exemplifies a new genre of **computational qualitative inquiry**—rigorous, transparent, and replicable yet anchored in human interpretive judgment. Empirically, it surfaces the current landscape of AI-awareness across subjects and grades, offering actionable insights for curriculum improvement. Practically, it delivers a usable toolkit for embedding disclosure, verification, and oversight without diluting pedagogical creativity.

As generative AI continues to evolve, educational integrity will increasingly depend not on detection after the fact but on **design foresight** embedded in every plan, rubric, and prompt. By empowering teachers to write AI-literate, ethically aware lesson plans, we lay the groundwork for **schools that are not merely future-ready but future-responsible**.

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